2023/HeQing

PORTFOLIO



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01

DISCLOSURE

The development of information technology as well as the arrival of big data society, has ushered in a series of problems, one of which is privacy. The project seeks to discuss the formation mechanism of social oppression caused by the game between enterprises and consumers in privacy issues, to intervene in the process of privacy game. By building a large, interactive building, visitors can experience the process of the privacy game, and be aware of the social oppression caused by the unequal rights of both sides, further to inspires people to pay attention to privacy issues in the data society and think about how to improve their rights in the game to reduce this oppression.



THEORY

This part illustrates the theory of social structures and analyses current privacy concerns in the context of the transition from disciplinary society to society of control.

Characteristic of exercise of power

Physical control

The willness of people

SOCIETY OF SOVEREIGNTY

Less effective. Usually exercise power after the crime occurred.

In society of sovereignty, the law presents the will of the king, the crime means the attack on the king, and the punishment is on behalf of the revenge of the king.

DISCIPLINARY SOCIETY

Lock people up in specific institutions, such as school, hospital, and prison.

Disciplinary society made use of three techniques: hierarchical observation, normalizing judgement and examination.

SOCIETY OF CONTROL

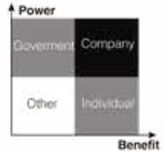
Does not require physical intervention, but through manipulation of flows of information.

Corporation replaces the factory, perpetual training tends to replace the school, and continuous control to replace the examination.

IDEA

Game Theory

Liu Ying and Chen Shuang use game theory to analysis the privacy problem between media and audience. Lisa Rajbhandari and Einar Arthur Snekkenes also use game theory to explain the privacy game between a user and an online bookstore. In real life, the inequality of power produces the social oppression at last.



Social Oppression

Inequality of Power

Formation of Stratification

Game Setting

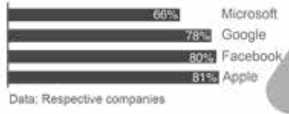
In the game of privacy protection between users and companies or companies, the goal of company or government is to earn more profits or regulate society by obtaining more data, while users expect to protect their data and privacy.



Panopticon

Panopticon, originated by Jeremy Bentham, is a conception in disciplinary society. In panopticon, a guard can observe all the inmates , while inmates never know wether or not they are being watched.

User data shared with US law enforcement, 2017



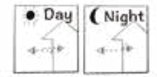
Privacy Protecttion

Privacy Attitude — Cultural difference / Individual difference

Unawareness of Users

Invisible Surveillance

Along with the change of the condition, the position from different side of the two-sided mirror will exchange, similar to the relation between onlookers and victims in data leakage.



Two-sided mirror

Manipulation of Data

Internet providers collect web browsing data and it to corporations and governments. Also, Microsoft, Google, Facebook, and Apple send personal data to governments without requiring the consent of users or even informing them about it.

Commodity — Agreement — Privacy

Privacy Leakage

Diverse Choices on Data Collection





Cookie, privacy policy

Data collection

Monitors

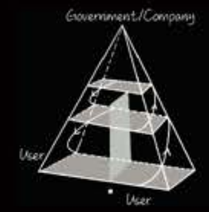
Massive methods have been used to collect data. Among these, cameras and monitors are able to show the process in a direct way.

Core / Important / Normal — **Data** — *extraction* / *record* — **Information** — Non-Personal Information / Personal Information

Private Information Replication

CONCEPT

To reveal the threats faced in our data society to ordinary users, such as data leak, and the social oppression, we design an exhibition space whose elements are inspired by the typical privacy data traps on the internet and the abstraction of related consequences.



Transformation of identity



Peep and being peeped



Attitude differentiation



Monitors everywhere



Exchange or loss of identity

IMMERSIVE ARCHITECTURE

Immersive architecture can lead users to be fully absorbed in an experience. In order to raise awareness of this disparity and intrinsic dilemma through the application of immersive architecture, this project designs an exhibition with the transformation of the privacy problem.

INTERACTIVE BODY DESIGN

Visible



Shredded paper with preprocessed privacy information.

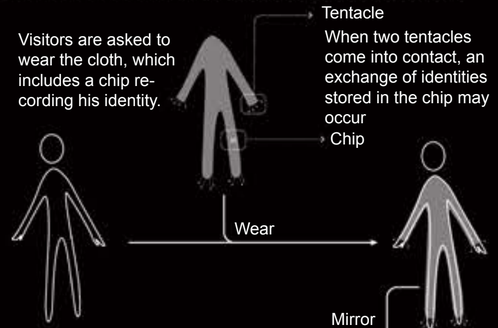
Invisible



Special fluorescent powder that can only be seen through a certain mirror in exhibition.

Privacy Leakage

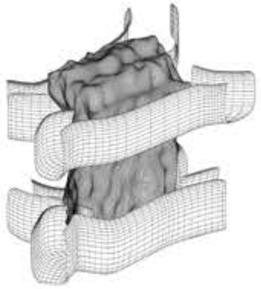
Visitors are asked to wear the cloth, which includes a chip recording his identity.

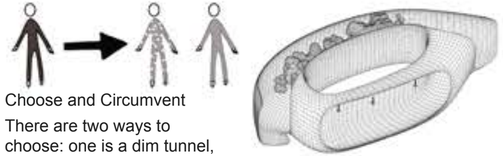


The clothes will continue to be stained with fluorescent substances and shredded paper during the visit, presenting a state of privacy leakage.

Mask Wall

One side of the wall, visitors need to climb the wall horizontally; another side, people walking by can see the faces of these visitors when their faces pass through some certain specific areas of wall.





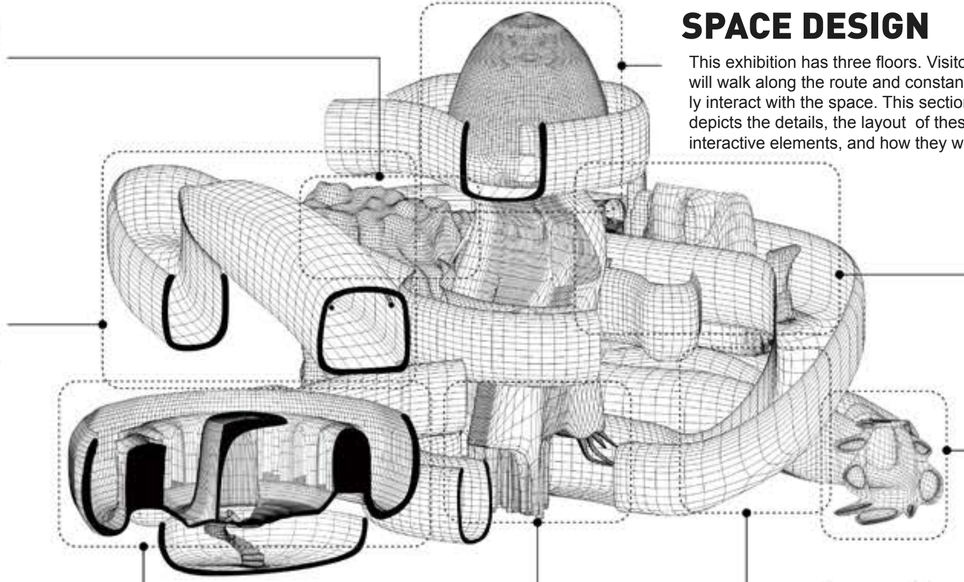
Choose and Circumvent

There are two ways to choose: one is a dim tunnel, which has an orbit above with some spotlights passing through. The light is full of invisible fluorescent powder; while another is a corridor full with small balls with paper scraps diffuse in the environment.

Mirror Rooms

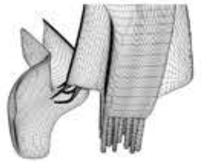
In each compartment, there are two mirrors, one mirror help to directly shows visitor the paper scraps messages on his garment, while another can make the fluorescent visible.





The Coral, Part 1

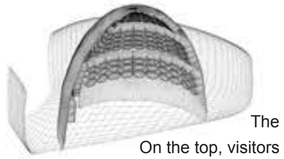
The Coral is the core of whole architecture. On each stair, there are several tentacles extend to the aisle. Through aisle, visitor can see some surveillance footages.





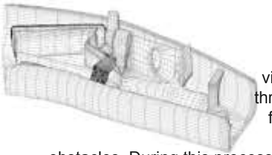
INTERACTIVE SPACE DESIGN

This exhibition has three floors. Visitors will walk along the route and constantly interact with the space. This section depicts the details, the layout of these interactive elements, and how they work.



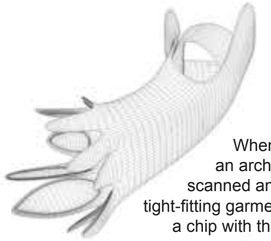
The Coral, Part 2

On the top, visitors can overlook the whole view of previous journey through a glass window. Then they can enter into a room full of screens.



Peep

In this space, visitors try to get through the roads full of deformed and squeezed obstacles. During this process, they will inadvertently see and be seen by other visitors.



Entrance

When going through an arch, visitors will be scanned and wrapped in a tight-fitting garment that inserted a chip with their own names.

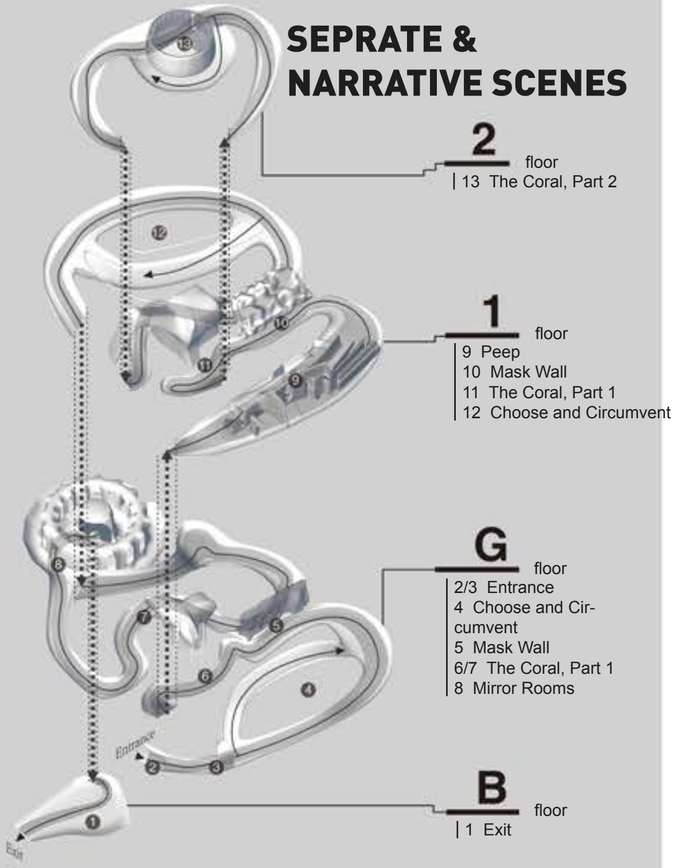


Exchange, the Staircase

Each staircase (exchanging stage), is made of a special erratic material that can keep the traces for a moment. Visitors will leave or stained with the fluorescent or paper scraps on them during climbing. And their chip information will be exchanged once two visitors have specific hand-foot, hand-hand or foot-foot contacts.



SEPRATE & NARRATIVE SCENES



2 floor
13 The Coral, Part 2

1 floor
9 Peep
10 Mask Wall
11 The Coral, Part 1
12 Choose and Circumvent

G floor
2/3 Entrance
4 Choose and Circumvent
5 Mask Wall
6/7 The Coral, Part 1
8 Mirror Rooms

B floor
1 Exit

Stage I - “Enter”

Helen is a seller working at the Apple. When he walked by a huge building, he saw a new building. When he went in, he found that it is actually a free gallery about technology. As he had nothing to do and was curious about it, he decided to have a visit.

2

Helen talked with the staff of this exhibition.

3

He passed through the laser door and was dressed in special clothes.

Stage II - “Visiting: being peeped”

Helen walked along the road and did the things signs asked to do. He felt that he got some stains on his clothes when walking through spaces with different settings. During whole process, he felt uncomfortable and a little bit scary.

4

Helen chose the lighter way and filled with paper cuts after getting into it.


When getting close, he saw some faces moving fast on the wall.

5

Through these tentacle-like devices, he saw some strange images like monitoring images.

6

When climbing, he found a laser quickly passing through his feet and hands covered with clothes.

9

To get through this, Helen discovered that others inevitably doing weird actions

Stage III - “Exchange”

Helen arrived at the center of the building, and he saw surveillance videos of different parts of the whole building. He realized that he had been completely under surveillance.
Then, his position in this visiting had changed.

13


Helen stood in a dark house and looked back whether he did some bad things in previous visiting.

Stage IV - “Visiting: peeping”

Helen participated in several similar installations in different ways. He felt that he was the man peeping on the visitors who had just come in.

12

Helen still chose the darker channel to prevent being caught by cameras.

5

Helen found that he became a mask face of the wall now.

Stage V - “Reflection”

The visit to the gallery was about to be over and Helen at last entered a mirror room. In the mirror, his robe, which he thought would not get too much paper cuts on, was covered in fluorescence.

8

He found his robe was filled with invisible fluorescence and paper cuts.

8

The door scanned him again, but this time the name that came out of the machine was not "Helen" but "Amy".

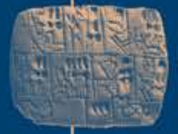
02

collecting

media technology

MUSEUM 4.X

Museum as an important media for people to attain their knowledge, has experienced four stages from collectors vision driven, to now the semantic museum. Its physical border has been expanding while knowledge border is fluctuating, which are hard to achieve the ethics of knowledges. Aware of this problem, this project breaks the old model of past museums and designs a pattern of Museum 4.x. An installation, designed to present this pattern, likes a clock to shows an invisible consensus among public, which can realize two-way interaction between museums and visitors, as well as interaction between visitors. The installation, however, is more than that and can be adapted to anyplace that the ethics of knowledges can be produced.

















3000BC 2500BC 2000BC 1500BC 1000BC 500BC 0 500 1000 1500 1600 1700 1800 1900 2000

Museum 1.0
collectors vision driven
530 BCE

Museum 2.0
cinteractive museum
1899

Museum 3.0
constructivist museum
1969

Museum 4.0
semantic museum
1987

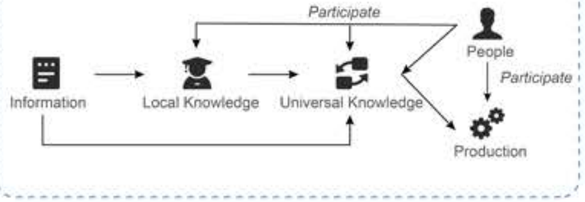
knowledge border

physical border

IDEAS

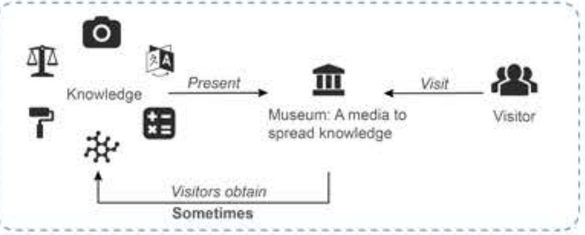
After entering the stage of museum 4.0, the knowledge border shrinks compared with previous stage. In order to deal with this shortcoming, we propose a museum 4.x based on the conception of ethics of knowledge.

Conception



Ethics of Knowledge

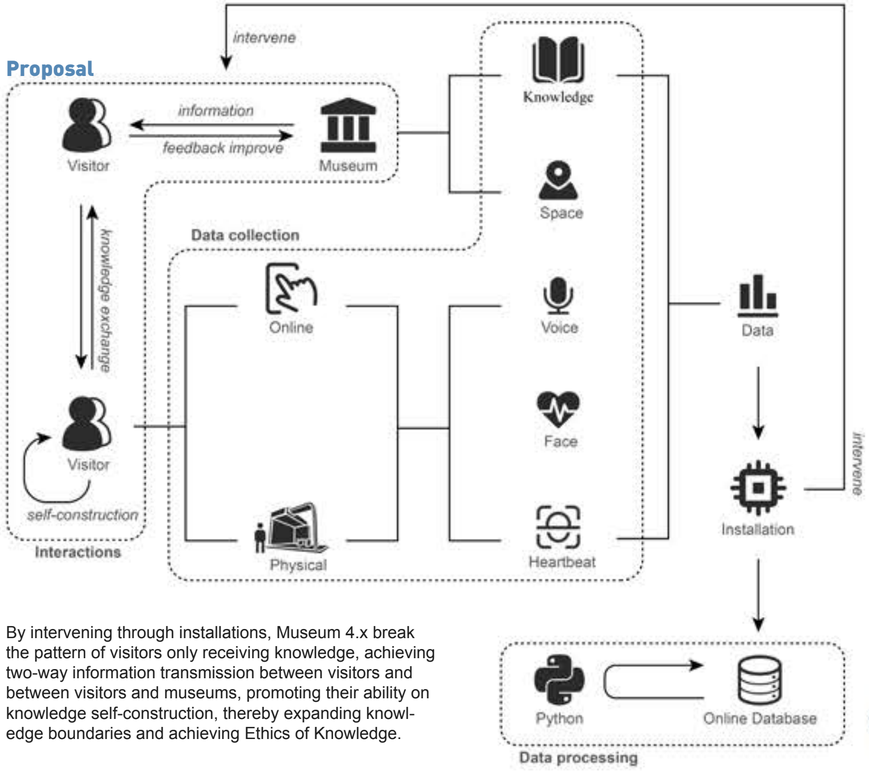
Not only care about the content of knowledge, but also focus on how knowledge is relationally created, how local knowledge can be transmuted into 'universal knowledge', and how the transaction and consumption of knowledge also monitors its subsequent production.



Creation Process: Ethics of Knowledge

This process describes that Ethics of Knowledge is created during obtaining knowledge from museum through visits.

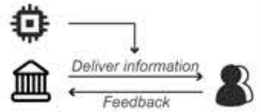
Proposal



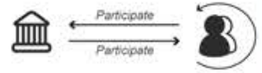
By intervening through installations, Museum 4.x break the pattern of visitors only receiving knowledge, achieving two-way information transmission between visitors and between visitors and museums, promoting their ability on knowledge self-construction, thereby expanding knowledge boundaries and achieving Ethics of Knowledge.

Museum 4.x

Museum 4.x not only helps visitors to absorb knowledge but also promote the creation of ethics of knowledge.



Visitors receive information and give feedbacks.



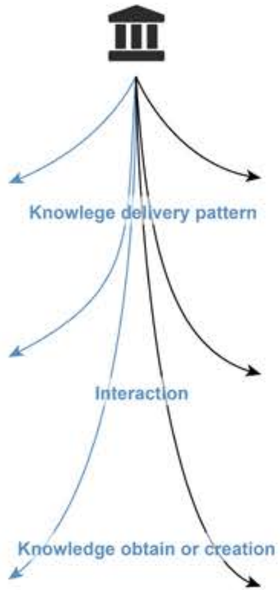
Interactions are depended on both museum and visitors.



Visitors can share own ideas and this helps to obtain or create knowledge

Receive knowledge positively.







Knowledge absorption turn from force to self-absorption.

Traditional Museum

Traditional museums pass the knowledge to visitors through exhibits and environments, but ignore how people will receive it.



Deliver information directly.



Participate in specific interactions.



Knowledge only comes from explanation like traditional teaching.

Receive knowledge passively.

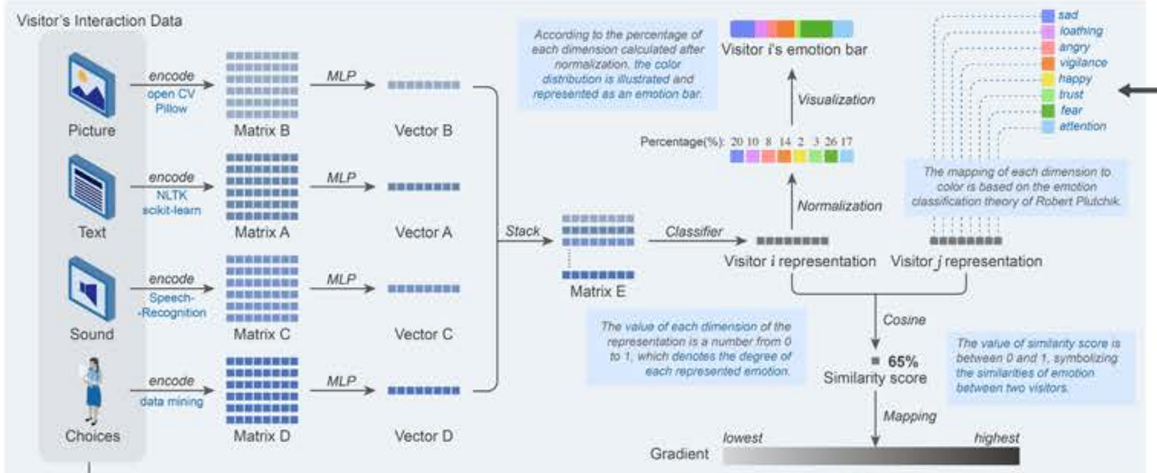
Comparison

The graph shows the difference between Museum 4.x and traditional museums.

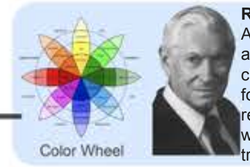
DATA WORKFLOW

This process depicts detailed process that is related to data, from data collection to data computation. Specially, the environment of data storage and computation is set on cloud. The data computation is based on Python, with different libraries in different tasks.

MUSEUM 4.X SYSTEM DESIGN Part One

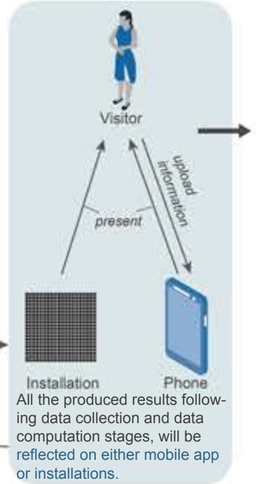


Hybrid information processing model structure



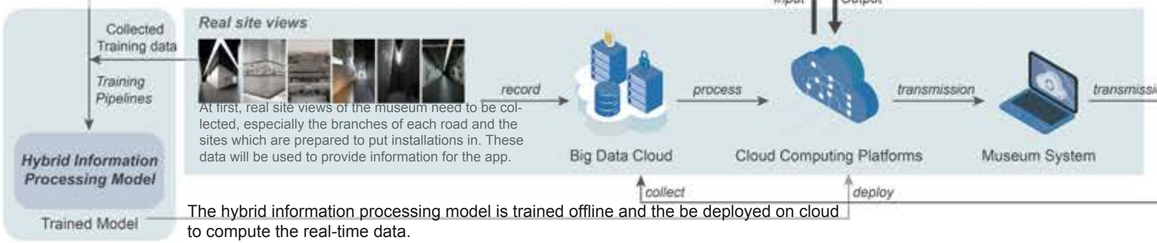
Robert Plutchik
A psychologist, proposed a psychoevolutionary classification approach for general emotional responses and created a wheel of emotions to illustrate different emotions.

Interactions with museum
All the interactions of visitors are collected for the calculation of the knowledge panel and other information shown in app.



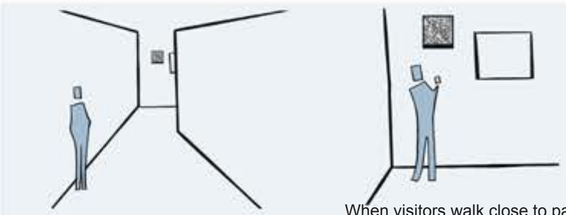
All the produced results following data collection and data computation stages, will be reflected on either mobile app or installations.

Offline model training



At first, real site views of the museum need to be collected, especially the branches of each road and the sites which are prepared to put installations in. These data will be used to provide information for the app.

The hybrid information processing model is trained offline and the be deployed on cloud to compute the real-time data.



Standing from a far distance, visitors can see the panel and be appealed by it.

When visitors walk close to panel, the bars on current exhibition spaces or items appears in their app.

Visualization Design

By clicking the specific bar of other people, visitors can compare their representations with it, which helps them reflect on these differences and think about the exhibitions.

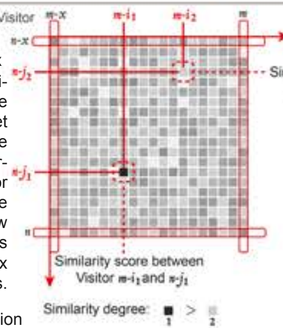
click to see specific information



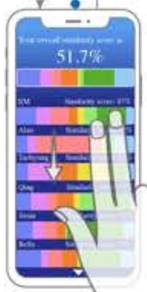
swipe down to see comparison

Emotion bar
The emotion bar of a visitor illustrates the proportion of different emotions in his/her whole emotion. The bars varies with different visitors and different exhibition items or spaces.

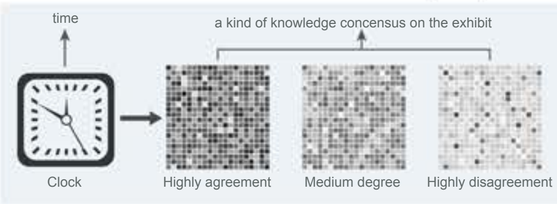
Similarity matrix
It involves the mutual similarity of x interactive visitors. The latest interactive visitors are set to default and displayed to the visitor. Since the current interactive visitor is the latest visitor to interact with the panel, the column n and row n show similarities between this visitor and former x visitors.



Bar comparison
The emotion bars of other visitors as well as the similarity score with each of them are shown on the APP. The visitor can compare their representations and think about the differences. Fingers can scroll for more visitors.



Knowledge Consensus Reconstruction

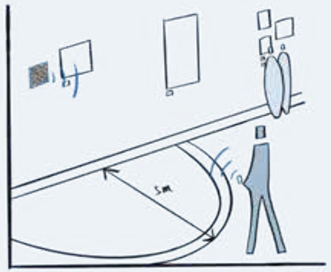


Humans created 'time' and reached a consensus on this concept, which is reflected through the clock. Likewise, we build a consensus of visitors' knowledge toward exhibits and the panel is the 'clock' in this reflection process.

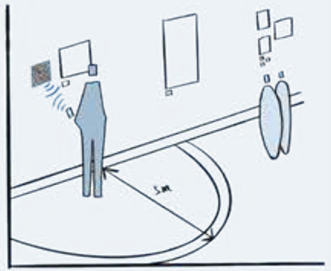
Visitors can see the overall attitude of previous visitors conveyed by the panel from a distance. The panel will leave a profound impression in their minds, even evoking some feelings of horror and fear, as it reveals something present in everyone but previously left unnamed, much like time.

MUSEUM 4.X SYSTEM DESIGN Part Two

2.1



2.2



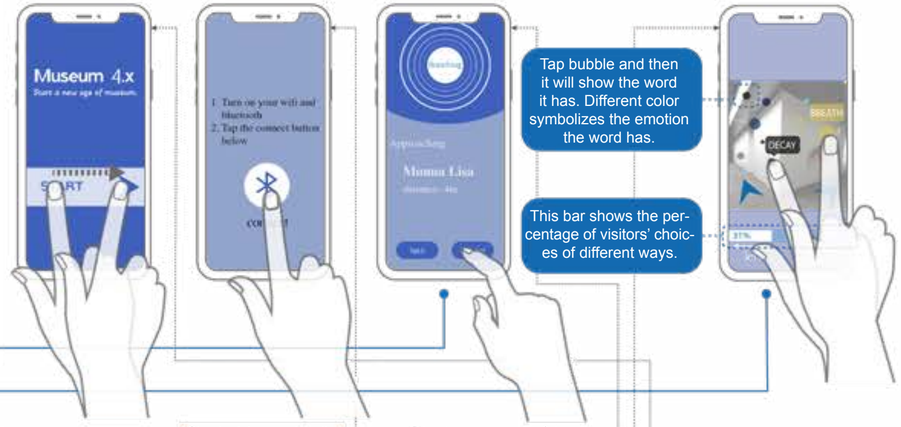
After using phone to connect museum system on app, phone will search for near installations in this museum. If visitor reach in the searching scope of the installation, then the installation will appear in the searching page and allow visitor to interact with.

Start

Connect to system

Search for near exhibits

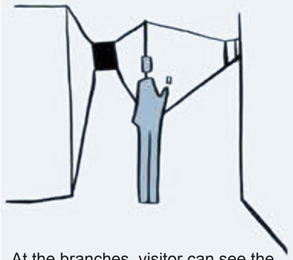
Tap color bubble for information hints



Tap bubble and then it will show the word it has. Different color symbolizes the emotion the word has.

This bar shows the percentage of visitors' choices of different ways.

3.1



At the branches, visitor can see the information of each branch on app. And then make decision on which way to go.

3.2



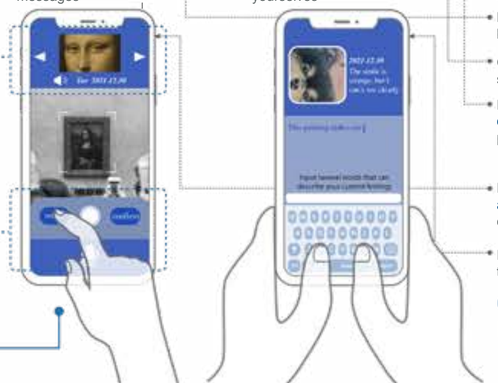
Visitors can interact with museums through online system and panels.

Check picture and audio messages

Look up messages & Input yourselves'

Tap these buttons to see the photos, and sounds recorded by previous visitors.

This part is used for visitors recording during visits, including views they saw and sounds they hear.



A. Swipe the screen to start.

B. Use phone to connect to our app by WIFI and bluetooth.

C. Bluetooth technology is used for searching nearest installations.

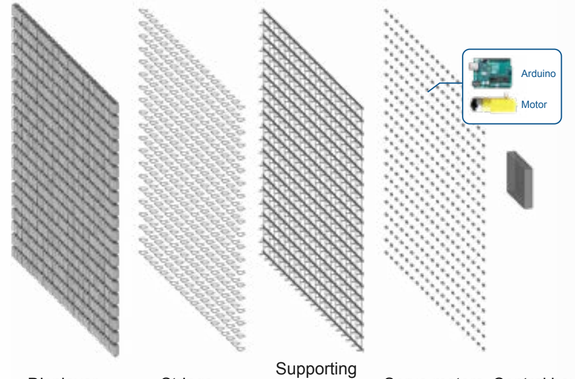
D. Through collected street view, data mining and machine learning to provide augmented reality.

E. Show other visitors recorded voice and picture, and allow visitor to record their own observations.

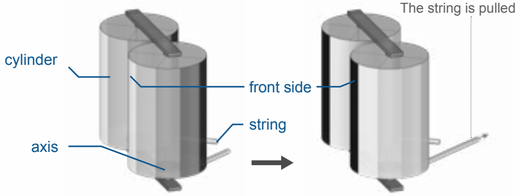
F. Use data extracted from database to realize interactions between visitors.

Online Platform UI and Interaction Design

Installation Exploded-view



Displayer | Strings | Supporting structure | Servo motors | Central box | Overall model



12:33:23 Similarity 32%-degree 3

12:33:59 Similarity 94%-degree 10

Each unit on the displayer consists of two cylinders, denoting a mutual similarity score. The surface of the cylinder is separated into 10 parts, showing 10 different degrees of similarity. Under the cylinder is an axis, and axes in the unit are connected with a string, which will pull the cuboid to rotate under the drive of the system and the motor while changing the color of the two cylinders facing outward.

EXPERIENCE MAP

In this part, we select a representative museum and scene as an examples to illustrate the practice use of our system.

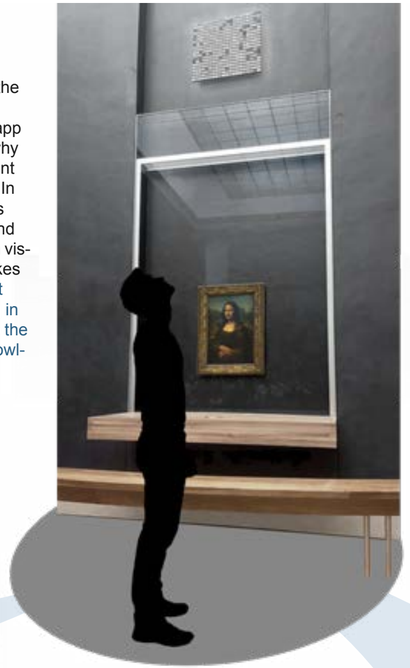


One day, Jk went into the Louvre museum to see Mona Lisa. He saw the poster which declared that Louvre had launched an app based on Museum 4.x system recently.



When waiting in the line, JK interacted with the app and find out that the panel changes all the time.

Visitors consider about the messages conveyed by both the panel and the app and then reflect about why people hold quite different thoughts on Mona Lisa. In this way, the interactions between the museum and visitors as well as within visitors happen, which makes visitors start to construct their own understanding in their minds and leads to the creation of Ethics of Knowledge.

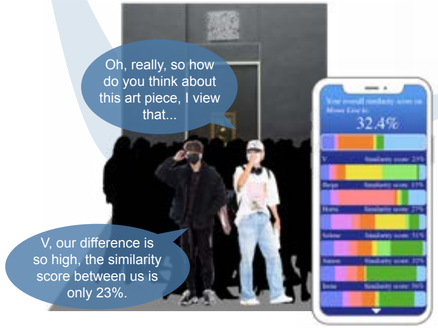




After a branch of the way, JK touched a yellow bubble which symbolized Da Vinci. Then, he turned right.



He entered the hall where Mona Lisa in and saw something hanging on the wall over this painting. He is confused and wonder what is that.



JK compared the self bar with other users. Then he found that his friend V had a very little similarity with him. They shared their views to try to figure out how it happened.

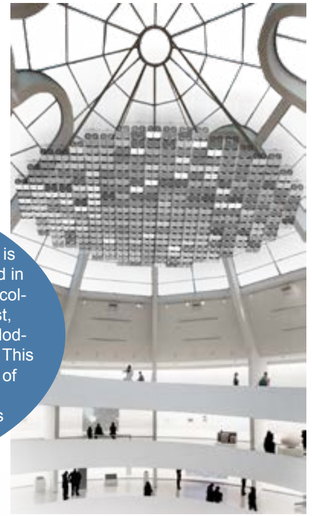
MORE THAN MUSEUMS

The pattern we propose above is not only suitable for different types of museums, such as art museums, history museums, science museums, etc., but also can be adapted to other places where knowledge sharing can take place, for instance, relics, building, school, or even the supermarkets.



Guggenheim museum

The Guggenheim Museum is an art museum, established in 1939, in New York City. Its collection is from Impressionist, Post-Impressionist, early Modern, and contemporary art. This panel is hung on the dome of the museum to showcase everyone's attitude towards the entire building.



The Great Wall of China

The Great Wall is a series of fortifications that were built in ancient times. Through climbing, people can see almost all the climbers share a similar feeling that are astounded by it.



03

“LEAP OF FAITH”

In the year 2090, the global environment deteriorated, temperatures increased, and the entire sky was directly exposed to the sun. Ultraviolet rays were particularly strong, and people began to fear and avoid the sun. Opposite to the inactive in physical word, the virtual word becomes the main place of living.



10 YEARS AGO, WORLD WAS SEPARATED INTO TWO PARTS, ONE IS NEO-WORLD, IN CHARGE OF NEW LIBERTY; ANOTHER IS OLD WORLD, NO ORDER, BUT EXISTS A PARTY CALLED THE GUARD.

C37, A SPY IN THE GUARD, WAS ORDERED TO INVESTIGATE THE TAMING MACHINES USED TO ANNEX THE OLD WORLD IN THE NEW WORLD.

THIS TIME YOU NEED TO GO ONTO THE NEW AGE, THEN .....

YES, SIR.

BE CAREFUL, WE HAVE LOST MANY PERSONNEL.

CHAPTER 1
INTO THE MONSTER

AFTER TWO MONTHS, I FINALLY ENTERED THIS BIG MONSTER.

THE LOBBY ROOM WAS FULL OF PEOPLE AND WAS A MESSY FREE-FOR-ALL.

PLEASE PUT ALL YOUR BELONGINGS ON THE SHELF...

THIS IS YOUR CHIP THAT RECORDS ALL YOUR INFORMATION, INCLUDING IDENTITY AND ASSETS.

I HANDED OVER ALL MY ASSETS, AND IT IMPLANTED A CHIP UNDER MY ARM TO RECORD IT.

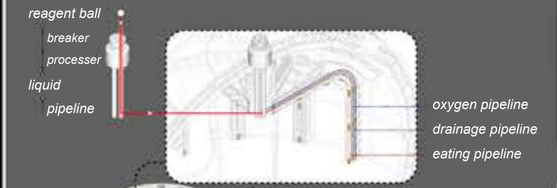
1 Apron
2 Lobby
3/4 Registration room
5 Hall 1
6 Stairs
7 Hall 2
8/9 Clean and spray room

ATTENTION PLEASE! .....

THEY BEGAN TO HINT THAT PEOPLE WERE GOING TO SEPARATE, AND EACH ONE WOULD ENTER THEIR OWN SEPARATE SPACE.

START SPRAY

I REMEMBER ...



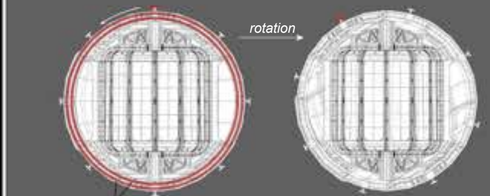
Substance Exchange

To provide the essential living demands for people to live in the cell, including oxygen, water and food.



Magnetic Plate

This is designed for virtual world living. After getting into the cell, resident will connect to virtual world automatically. The magnetic plate can help resident simulate all actions in the physical world.



Moving System

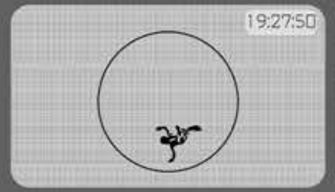
Moving system can make cells move in the sophisticated tracks in New Age. The Two-layer moveable shells ensures that a 360-degree rotation of the outer shell will not impact the lives of the individuals within the inner shell.

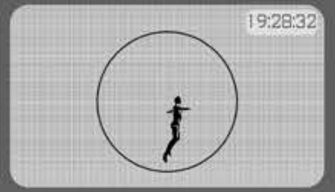






















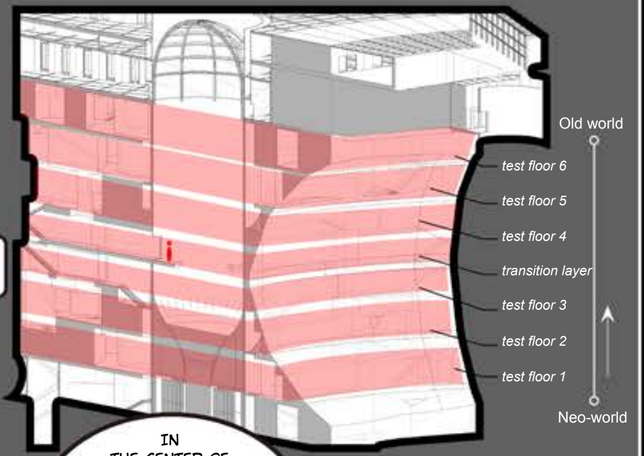


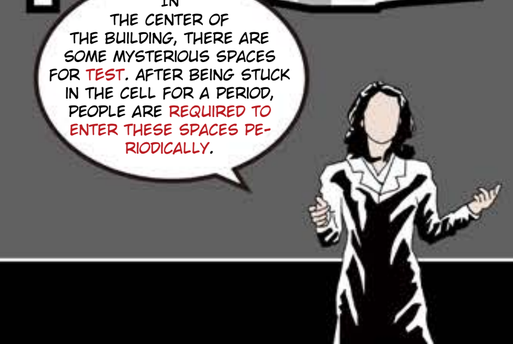




Test Spaces

The higher the floor is, the more closer to old-world the setting is. This is designed to test the extent to of the tested people are accustomed to neo-world.





CHAPTER 2 LIVING IN THE CELL





FINALLY COME OUT FROM THE CELL. BUT WHAT PLACE IS THIS AND WHY ARE THESE FACILITIES SO SIMILAR TO THE OLD WORLD.



SO FOGGY. IT SEEMS THAT THERE IS NO WAY LEFT, ONLY A CLIFF AHEAD.

!!



A CLUE?

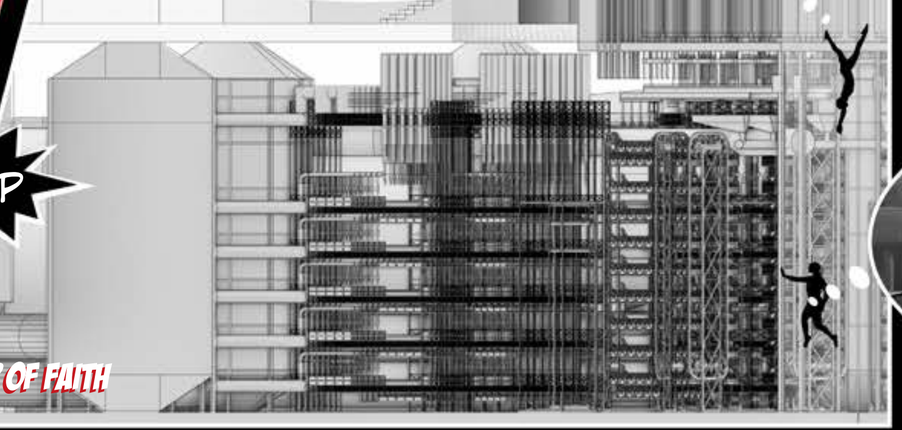


JUMP

CHAPTER 3
FACTORY: TAKE A LEAP OF FAITH

IT SEEMS LIKE A DOOR. OH, MY FEET CAN GO THROUGH IT.





CHAPTER 4 UNDER WATER

WHERE AM I...

I'M UNDERWATER...

IT SEEMS LIKE A FACTORY OUTSIDE. THIS MIGHT BE A REAGENT PRODUCTION FACTORY FOR NEW WORLD. HOWEVER, THE DECODER THEY ADDED IS VERY FAMILIAR.

WAIT, ..., IT SEEMS TO BE THE CUSTOMIZED DECODER FOR THE GUARD.

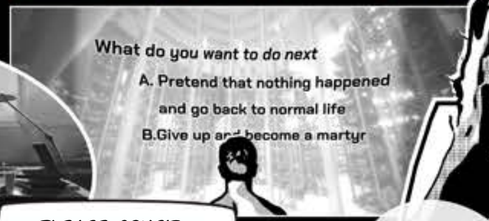
BUT WHY THEY USE THIS...



THIS DECODER IS A SPECIALLY CUSTOMIZED DECODER THAT ONLY OUR ORGANIZATION OWNS.



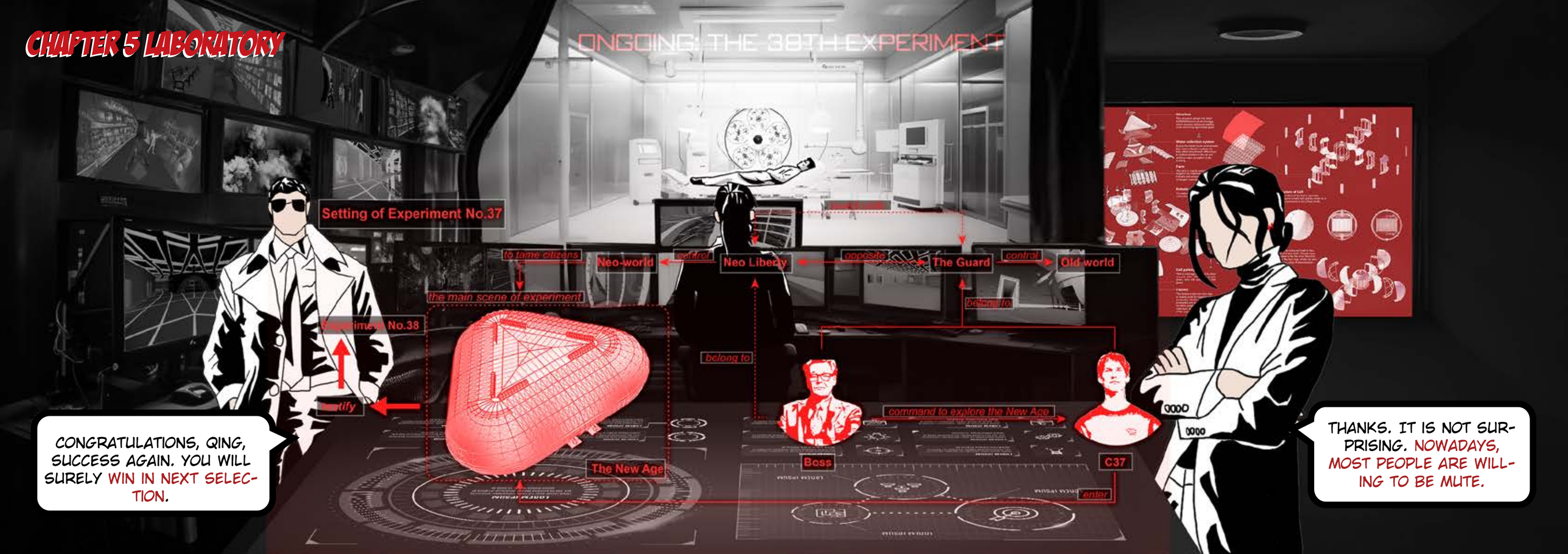
C37, WELCOME, WE HAVE BEEN WAITING FOR YOU FOR SO LONG. THESE ARE YOUR PREDECESSORS. NOW, IT'S YOUR TIME TO CHOOSE.



PLEASE CONSIDER CAREFULLY AND CHOOSE ONE OPTION



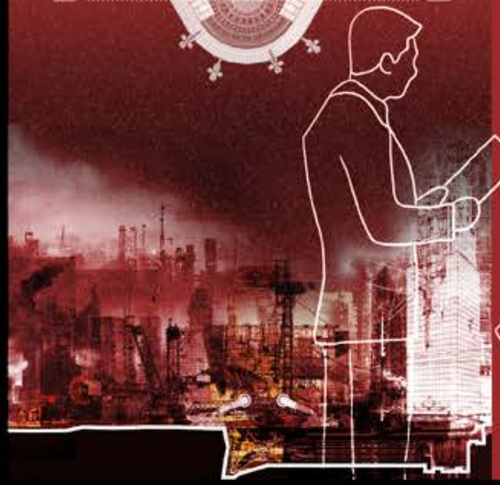
SO WHO AM I...
WHAT AM I FIGHTING FOR...
IS THIS HOW FREEDOM ACHIEVE...
SHOULD I IGNORE THE TRUTH...
DOES FREEDOM EXIST ???!!!


CHAPTER 5 LABORATORY
ONGOING: THE 38TH EXPERIMENT
Setting of Experiment No.37
to tame citizens
Neo-world
control
Neo Liberty
opposite
The Guard
control
Old world
the main scene of experiment
belong to
Experiment No.38
belong to
notify
command to explore the New Age
The New Age
Boss
C37
enter
CONGRATULATIONS, QING, SUCCESS AGAIN. YOU WILL SURELY WIN IN NEXT SELECTION.
THANKS. IT IS NOT SURPRISING. NOWADAYS, MOST PEOPLE ARE WILLING TO BE MUTE.



Top







Front



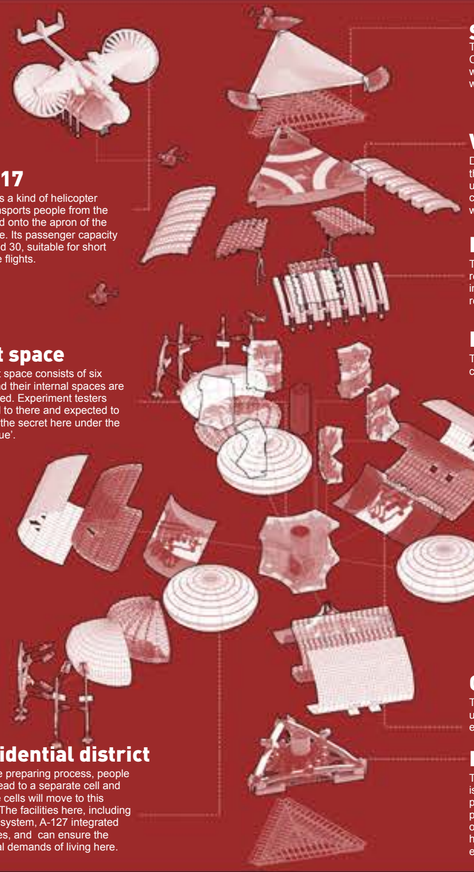
Section AA'



Section BB'



Section CC'



AZ317

AZ317 is a kind of helicopter that transports people from the old world onto the apron of the New Age. Its passenger capacity is around 30, suitable for short distance flights.

Test space

The test space consists of six parts and their internal spaces are connected. Experiment testers are lead to there and expected to explore the secret here under the given 'cue'.

Residential district

After the preparing process, people will be lead to a separate cell and then the cells will move to this district. The facilities here, including C3-123 system, A-127 integrated machines, and can ensure the essential demands of living here.

Structure

This structure adopts the latest CASPAK framework technology, which ensures structural stability while achieving lightweight goals.

Water collection system

Due to the harsh future environment, this water collection system can fully utilize temperature differences to collect moisture in the air and achieve water circulation in the building.

Farm

The farm is mainly used to provide reagent raw materials to the industry and ensure the progress of reagent manufacturing.

Robotic arm

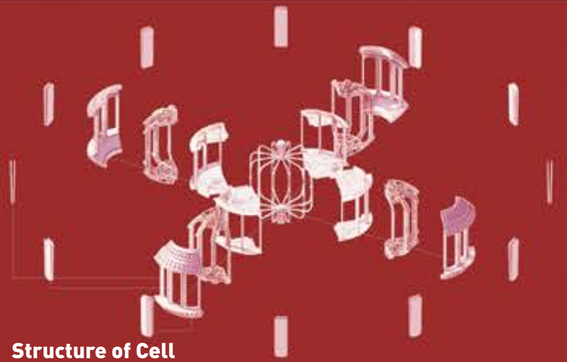
To realize the automaticity of the cultivation and harvest of crops.

Cell parking-lot

This is a storage and used to place unused cells. When new people enter, cells will come out of this place.

Factory

The factory inside the New Age is mainly used for reagent ball production. All the process of the production are automatic so that no other people are required to work here. It is also the final stage of the experimenter's testing.



Structure of Cell

The structure of the ball is specially designed to enable anti gravity mode for a better experience in the virtual world.



Cell

Cell is the living unit both in New Age and Neo-world. People need to adapt to the life when they first enter the New Age, which can also be seen a kind of domestication.



Cell shell

There are many track fixing components on the outer shell of the cell, which help the cell rotate 360 degrees without obstruction.

04

EMOTION SPACE SELF-CONSTRUCTION

Living in a fast-paced life with high competition, I noticed that people are heavily exposed to public space, which can squeeze people's private space. These two things lead to a contradiction between negative emotions the life brings and the insufficiency of private spaces to deal with these emotions.

In this project, we intervene in the process of emotion generation by constructing a systematic theory of emotional space, ultimately to help people deal with negative emotions in public spaces. We design a game to simulate the process of emotional space self-construction and a wearable installation to present the interactions with emotional assemblages.



BACKGROUND

Nowadays, we spend most of our time in public spaces, such as offices, schools, subway stations, and even the places many people live in are shared apartments.

If suppression is not a good idea to deal with our negative emotions, what could be? We can see that one of the prominent problem is the lack of private space, so in this project, we propose an idea to construct an emotional space, which can extend our private space, so that we will not feel squeezed and help us to deal with our emotions.



The long exposure time in public places, makes us feel squeezed as we don't have a private space to deal with our emotions, especially negative emotions. Therefore, suppression becomes the main method for emotion regulation. However, according to the research, suppression can lead to some bad consequences, such as a decrease in memory ability, an increase in blood pressure, and worse mood repair.

CONCEPTION

Emotion Assemblages

It is an aggregation of any thing, for example, a piece of music, a dog, the sky you see, that can be accepted by yourself under current emotional circumstance.



The man is so happy that the sun, the music in his headphones, the chirping of cicadas, and even the roar of cars can fit into this joyful atmosphere.



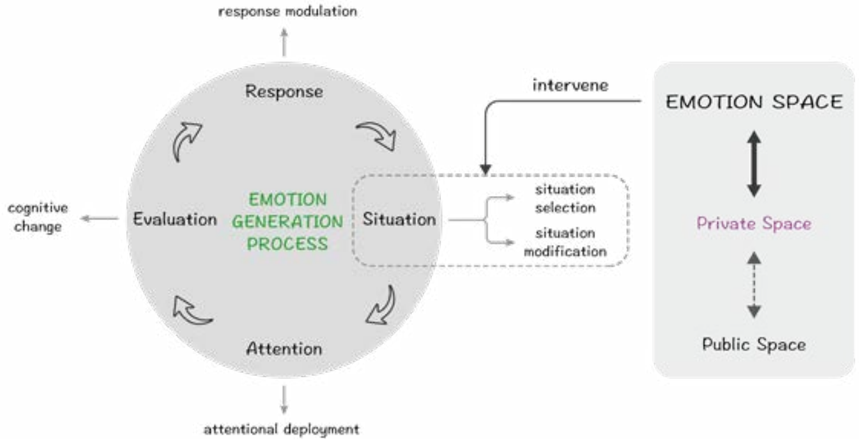
The people who is crying, walks on the street with his dog. He listens to a song named Rainy Days, the dark clouds always company him.

Emotion Space

Emotion space is a cognitive form and can be seen as an extension of private space, as opposed to physical form. It is composed of emotion assemblages.

Objective

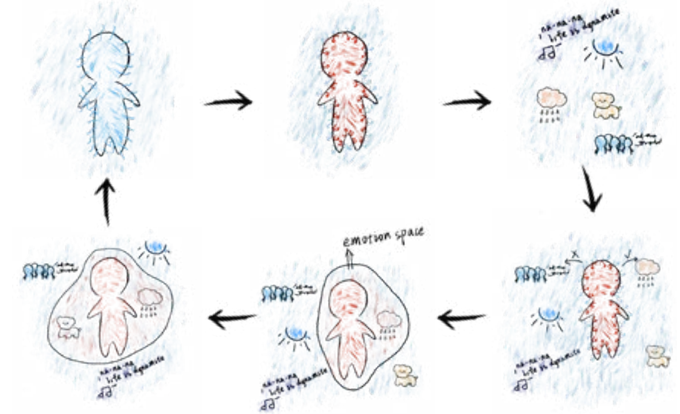
The objective of emotional space is to intervene the process of emotion generation in situation stage, then finally helps people deal with their negative emotions in public spaces.



Emotion Generation Process

Emotion generation process, proposed by James J. Gross, consists of four stages, including situation, attention, evaluation and response. It describes how emotions are generated: in some situations, people pay attention to certain features and evaluate the impact on themselves. The evaluation then leads to a coordinated set of responses that change their feelings and behaviour.

Inspired by the meditation tradition of using skin to breathe in buddhism, we assume body is the contain of our private space and skin is the border. When people are in negative emotions and feel squeezed with insufficient private space, they construct emotional space to extend private space.

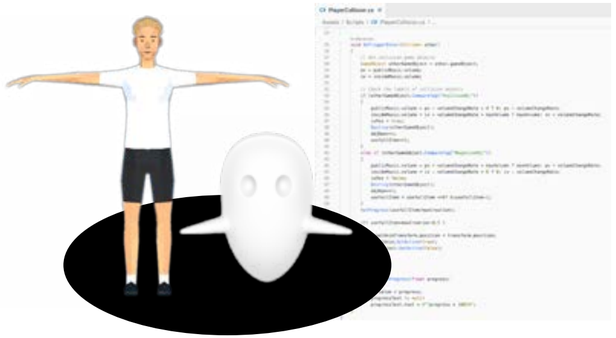


EMOTION SPACE CONSTRUCTION PROCESS

The construction process of emotional space is an active process. At first, treat an individual as a primitive emotional assemblage. Then the individual look for each surrounding element and judges whether he expects it appear around him under this circumstance. The accepted elements together with the primitive emotional assemblage combine into a larger emotion assemblage.

At normal state, our internal body can exchange emotion with public space, which can make our emotion achieve a dynamic balance. We can easily receive and response emotions that surroundings bring to us. When people in negative emotion, the exchange process can not work well. The internal bad emotion is dashing on body, while the emotion outside also influence people and make them feel uncomfortable. They requests more private space. After people success to construct an emotion space, their private spaces extend.

GAME DESIGN



The game is developed on Unity and illustrates a scene to show when people need and how to create an emotion space.

The more information about this project can be found on github:
https://github.com/HqSelene/EmotionSpaceCreation.git

The game can be played on website:
https://hqselene.github.io/EmotionSpaceCreation/

GAME SCENE





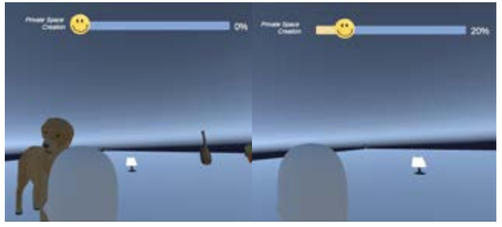
Selene is walking on the street. She feels blue today.



Hearing the noisy sounds, such as cars beeping, people speaking, suddenly she felt very devastated.



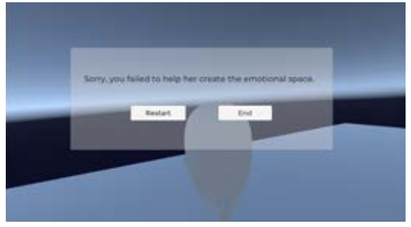
Now, you need to use keyboard and mouse to control the ghost of Selene to help her construct the emotional space.



You meet the dog and are ready to collect it. After colliding with the dog, you successfully help Selene's space grabbing process advance by 20%.



Then you meet the car and hope it can also be helpful. Unluckily, the process falls back and it is seems like that she does not very interested in car when she feels upset.



If you fail then the game will ask you whether you want to restart.

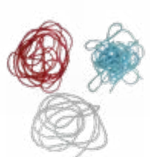


When this page jumps out, it means you successfully help Selene create the emotional space so that she can recover from negative emotions.

WEARABLE INSTALLATION DESIGN

This section shows the theory in depth through a wearable installation and performances, using our bodies to connect our emotional spaces physically. Hence, this part develops my affordances to identify other emotional assemblages and the process has been documented through a film.

MATERIALS


Mesh tube


Gloves


Bodysuit


Needle and thread


LED light


Mask


Iron wire


White cloth

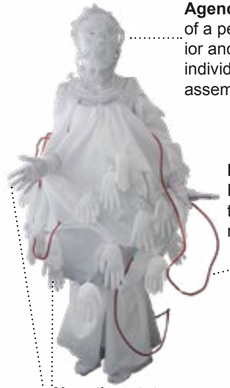

Transparent tape


Plier


Vise


Scissors


Hot melt glue gun



Agency: The subject of a person's behavior and a meta unit of individual emotional assemblage.

Emotion substance: Red denotes negative status while blue represents positive.

Negative status

Emotion feeler: Feelers are on the emotion border. Agency use feelers to interact with outsides.

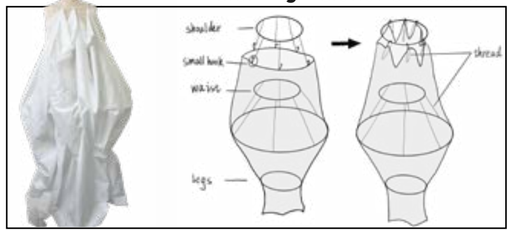
Emotion border: It is elastic and isolates the emotional space from the outside world.



Positive status

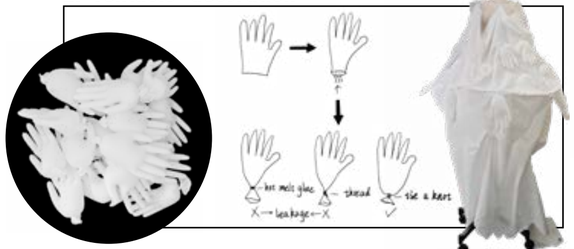
INSTALLATION DESIGN PROCESS

1.Clothing installation skeleton



Use wire and cloth to make the structure and shape of clothes.

2."Tentacle" decoration





Inflate the gloves and decorate them on the device.

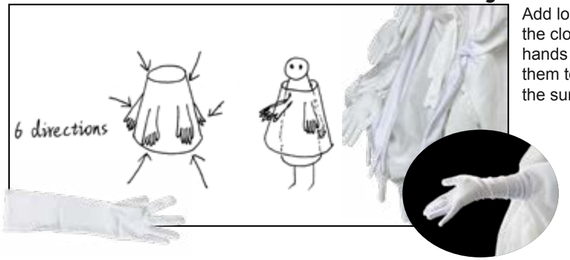
3.Shaping mesh tubes





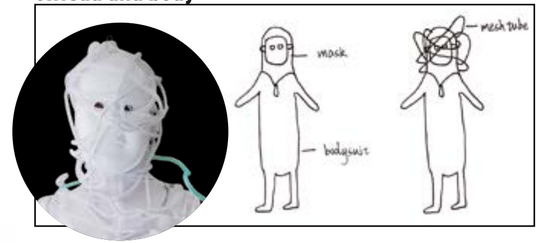
Input a long wire into a mesh tube, and then warp the wire to shape the mesh tube.

4.Gloves design



Add long gloves on the clothes so that hands can get into them to interact with the surroundings.

5.Head and body



Use a body suit and a mask to cover all the skins and mesh tubes to decorate the head part.

6.Stage property



Similar to the design principle of the main clothes, use iron wire to construct different stage properties and insert the LED light into them.

PERFORMANCE
—EMOTIONAL "MONSTER"

I wore the designed installation clothes and filmed a video to show the process of gathering other things into an emotional assemblage. The full video can be found at:

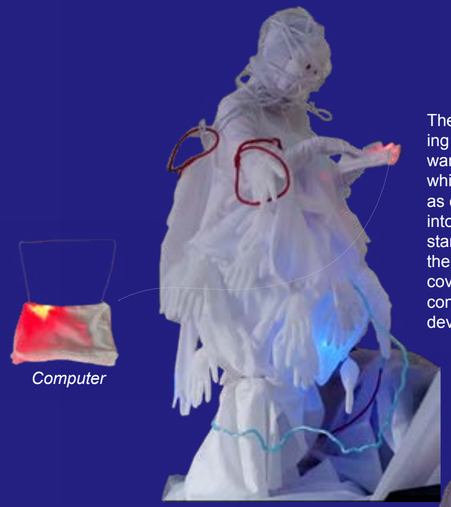
https://youtu.be/fM26rF-7J4s.



In this setting, several things are put into different places of this setting and white cloth, which denotes the emotional space of the agency, takes up only a very small space at first.



Noisy people



Computer



Book

The agency roams in space, slowly exploring and touching things of different colors. Blue represents things that it wants to approach, such as desserts, dogs, and guitars, while red represents things it dislikes at that time, such as computers, balls, and noisy people. It puts blue things into its huge body, making the emotion assemblage constantly expand. As the scope of exploration increases and the number of blue objects in the agency increases, the coverage of the white cloth (symbolizing emotional space) continues to increase, and the previously red coils on the device gradually turn blue.



Donut



Dog



Guitar



Birthday hat





05

ANALYSIS OF MUSICAL INFLUENCE

This is a team project, recognized as Finalist in the MCM/ICM 2020 competition, focused on applying mathematical models for data analysis. With a particular emphasis on the evolution of music and its influencing factors, the team, consisting of He Qing, Hu Ruiqing, and Yin Liang, collectively explored the provided dataset. My primary responsibilities in the project included mathematical model constructions and graphical representation by programming, contributing significantly to our success in data analysis of music evolution.

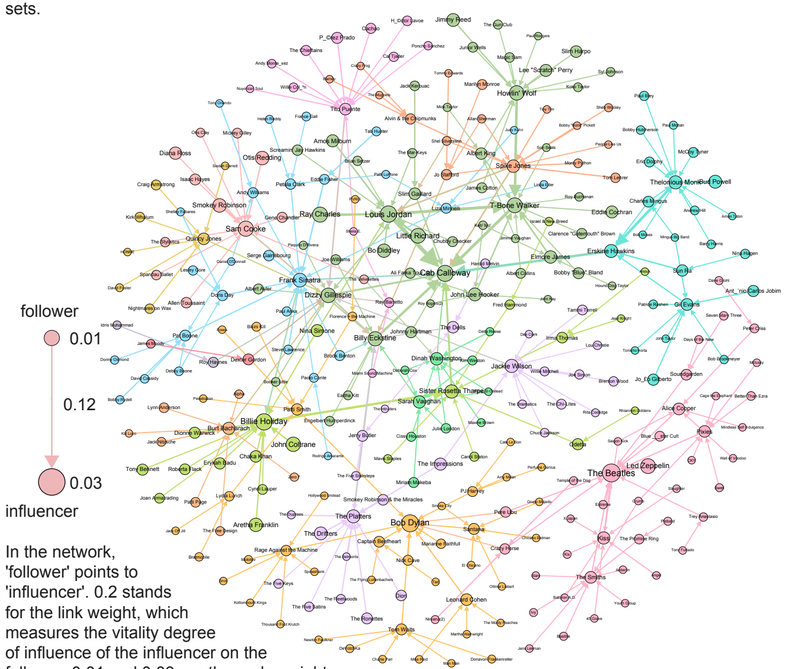
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https://github.com/HqSelene/2021MCM-ICM

Musician Influence Network

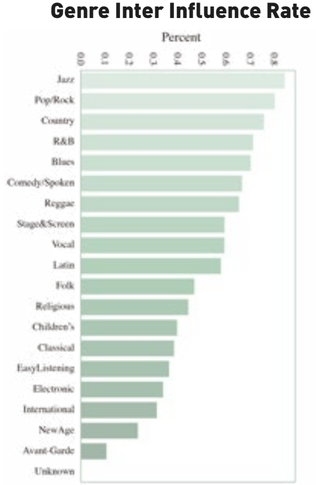
Musical influence networks are constructed based on the relations between influencers and followers, and describe musical influences by employing PageRank algorithm and exploring subsets.



In the network, 'follower' points to 'influencer'. 0.2 stands for the link weight, which measures the vitality degree of influence of the influencer on the follower. 0.01 and 0.03 are the node weights of follower and influence respectively and measure the influence of a musician.

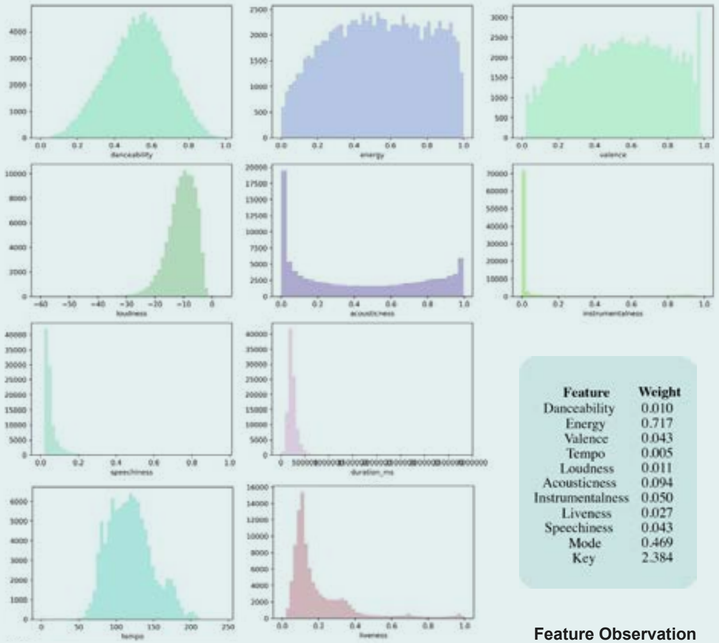
INFLUENCE ANALYSIS

Genre Inter Influence Rate



Mainstream music genres have numerous listeners and long history, which have formed their own unique and styles, so it is mainly influenced by its own music genre; and relatively niche music genres generally have not formed its own stable style. So it will be greatly affected by other genres.

MUSIC SIMILARITY CONSTUCTION



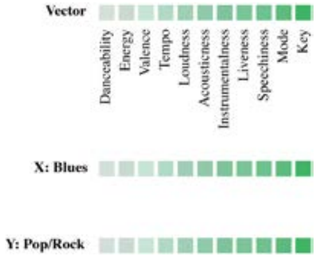
Feature	Weight
Danceability	0.010
Energy	0.717
Valence	0.043
Tempo	0.005
Loudness	0.011
Acousticness	0.094
Instrumentalness	0.050
Liveness	0.027
Speechiness	0.043
Mode	0.469
Key	2.384

Feature Observation

These features are characteristics of music and type of vocals. Some features show significant differences between music while other features witness marginal. Through the weight calculated by Entropy Weight Method, key and danceability are the most and least important in defining different music styles.

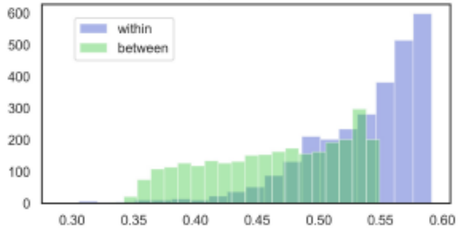
Similarities Between and Within Genres

A vector is a representation of an artist, a piece of music or a genre, and each dimension stands for a specific feature. The vector of the genre is the mean value of all the music pieces belonging to it.



Genre	*Genre_id*
Avant-Garde	1
Blues	2
Children's	3
Classical	4
Comedy/Spoken	5
Country	6
EasyListening	7
Electronic	8
Folk	9
International	10
Jazz	11
Latin	12
NewAge	13
Pop/Rock	14
R&B	15
Reggae	16
Religious	17
Stage&Screen	18
Unknown	19
Vocal	20

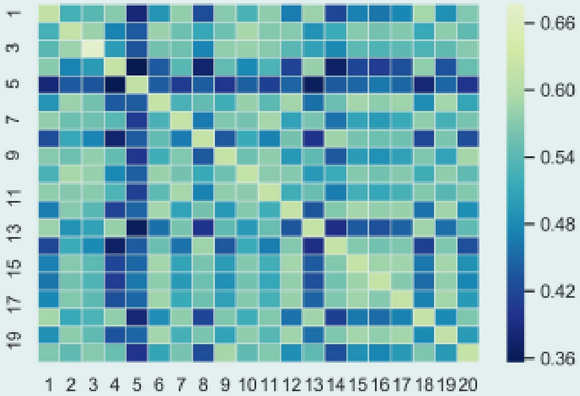
Artists similarities within Pop/Rock or between it with other genres



The similarity of artists within genre is obviously bigger than the similarity between genres.

The mean value of the similarity between each artist in the popular music genre and the geometric center value of all music genre is 0.537, while the mean value of the similarity between each artist in the popular music genre and the geometric center value of all music genres is 0.465.

Genres mutual similarities



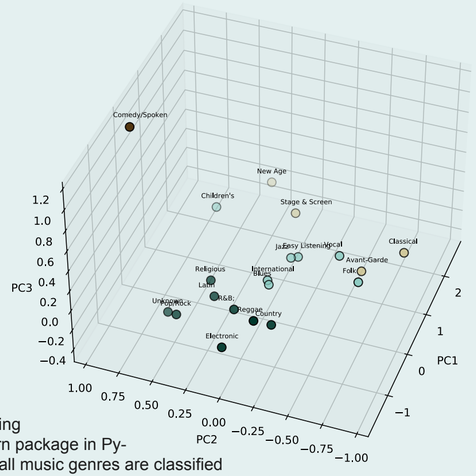
This heat map shows the similarities between every two genres calculated on mean values of them as well as mean similarities of musics within each genre. **The lighter the color, the higher the similarity**, vice versa. It is noted that the **similarities within genres is much bigger than the similarities between genres**.

MUSIC GENRE CLASSIFICATION

In order to further explore the differences and relationships between different music genres, we decide to use **Hierarchical Clustering Analysis** to gather similar music genres and separate the significantly different genres.

Classifications of Groups
R&B, Pop/Rock, Electronic, Latin, Country, Religious, Reggae
Comedy/Spoken
Jazz,EasyListening, Vocal, Folk, International, Blues, Children's
Stage&Screen, Avant-Grade, NewAge, Classicle

Principal Component Analysis clustering diagram



By using sklearn package in Python, all music genres are classified into 4 groups. In order to make the clustering results more intuitive, we used Principal Component Analysis (PCA) to reduce the dimensionality of the data to display the clustering results.

What Distinguish a Genre?

In Hierarchical Clustering Analysis, we choose the Complete Linkage to calculate the distance between two brands to maximize the differences between groups. To search characteristics that distinguish genres, we replace the four groups with geometric centers of each groups, then calculate the difference of each characteristic in four group.

By observing the result, we find that in instrumentalness and speechiness two characteristics gap between those four groups is significant. We believe the instrumentalness and speechiness are the key features to classify the genres.

Features	D_j
acousticness	1.414725
danceability	0.594540
energy	1.390823
instrumentalness	16.288928
key	0.232365
kind	1.500000
liveness	2.525010
loudness	-0.497463
mode	0.106799
popularity	0.515284
speechiness	9.101242
tempo	0.125223
valence	1.252960

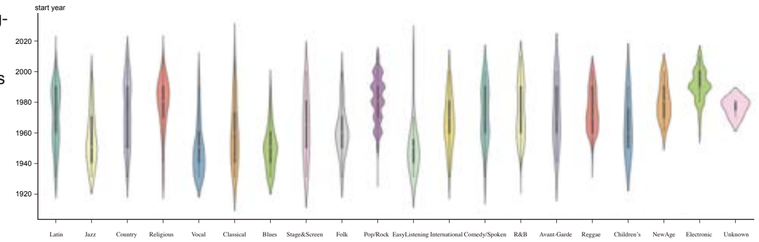
Who are the Dynamic Influencers?

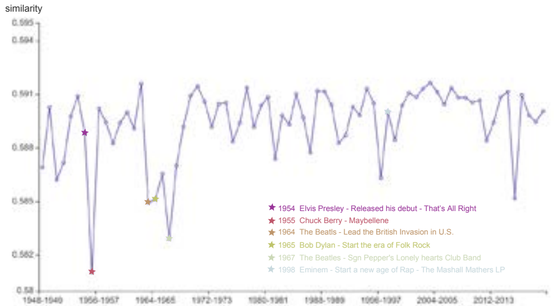
Ranking	By PR (1)	By DPR (1)	The value of
1	TheBeatles	TheBeatles(+0)	0.0044
2	BobDylan	BobDylan(+0)	0.0025
3	TheRollingStones	HankWilliams(+1)	0.0024
4	HankWilliam	BillieHoliday(+2)	0.0022
5	LouisJordan	LouisJordan(+0)	0.0019
6	BillieHoliday	TheRollingStones(-3)	0.0019
7	LesterYoung	WoodyGuthrie(+8)	0.0018
8	CabCalloway	LesterYoung(-1)	0.0017
9	DavidBowie	CabCalloway(-1)	0.0017
10	RoyAcuff	RoyAcuff(-2)	0.0016
11	ChuckBerry	DavidBowie(-2)	0.0016
12	JimiHendrix	MilesDavis(+5)	0.0016
13	ElvisPresley	JamesBrown(+1)	0.0015
14	JamesBrown	JimiHendrix(-2)	0.0014
15	WoodyGuthrie	MarvinGaye(+10)	0.0014

This table illustrates the importance ranking of influencers in the PageRank algorithm and dynamic PageRank method, which is modified based on the previous algorithm by introducing dynamic features into the previous algorithm.

Music Revolution

In order to study the changes of music genres over time, we have counted the number of artists of various music genres in different years, and draw the box plots based on the statistical results, which shows the change of popular music genres in different periods.





We use time series to calculate the similarity between years, and connect it with events (Figure12). During 1954-1956 and 1964-1967, similarity had significant changes and maintained a low position. In 1954 and 1955, Elvis Presley and Chuck Berry ushered in the era of rock and roll. In 1964, Beatles' concert in the United States led the British invasion, while Bob Dylan created a new era of folk rock. The cultural integration of music makes people constantly in- novate in this field, and the low similarity between years also shows that artists are constantly trying new styles in music. Since the 1990s, with the continuous development of society, the forms have gradually diversified and minority music gradually rose. While at the same time, due to the development of network and the influence of mainstream media, music always has a main line of development, leading to fluctuations after the 1990s.

06

A DATA PORTRAIT OF 1183 HELP SEEKERS



NOT THE WEEK, BUT EVERYONE

On 23 January, 2020, Wuhan started its lockdown which lasted for 76 days. According to statistics, there were about 9 million people in Wuhan during lockdown, and 5 million people had fled before it. Wuhan was the initial place of an outbreak of the COVID-19, hence that the government imposed the lockdown in an effort to quarantine this place.

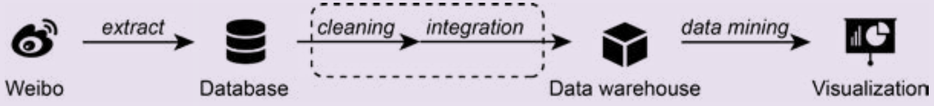
However, due to the immature strategies in face of the epidemic, there were a lot of patients could not receive treatments on time. The number of dead people increased everyday, the things we need to remember was much more than that. As WHO Director-General's stated, "We must remember that these are people, not numbers".

Therefore, a lab in our university collected data from Weibo (one of China's largest social media platforms) for the period from February 3, 2020, to February 10, 2020. Finally, we got 1183 effective information of these help seekers. Based on these data and some other public data, I visualized these data, generated a portrait for these help seekers and then made analysis for all the graph I made.



DATA PREPARATION

Before analysis and mining, we need to preprocess the raw data extracted from Weibo to deal with the problem that exists in raw data. This is called data preparation stage.



Cleaning

In reality, our data is usually 'dirty'. For example, the data we obtain can be wrong, inconsistent, missing, or ambiguous. If you input the dirty data into the model, then the result will also be like garbage. Which lead to 'Garbage In, Garbage Out'. In this stage, we use different ways to solve different types of 'dirty data'.



Ambiguity of data: Ambiguity means that the same texts have different references in various situations. For instance, 'Apple' can refer to a kind of fruit as well as a brand. 'Michael Jordan' is a famous basketball star but also can be a scholar studied in computer science and statistics.

Missing data

	name	age	city	addr	user	pubdate	pubtime
49	0	NaN	武汉	硚口区汉口春天	2252.0	2020/2/8	0 days 21:19:33.000000000
130	0	NaN	武汉	硚口区汉口春天	2198.0	2020/2/8	0 days 21:37:55.000000000
402	0	NaN	武汉	硚口区汉口春天	2385.0	2020/2/8	0 days 22:44:56.000000000

Missing data: This is a common problem happened in daily lives. There are several ways to replenish them, such as average value, approximation or Bayesian estimates.

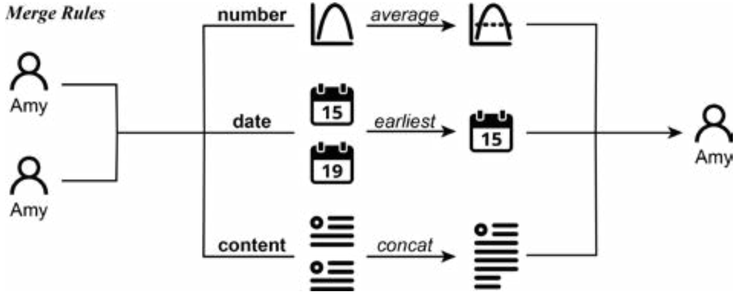
The count of null value in each categories of raw data.

name	0
age	120
city	17
user	6
pubdate	0
pubtime	0
content	2401
heading	950
link_lianjia_score	0
user_gender	0
user_province	0
user_city	0



Integration

After cleaning, there are 4961 data left. Through observation, however, some of the data have repeated information. Hence in this stage of integration, we merge the information by constructing several rules.




```
weibodata.content=weibodata[['content']].fillna( "N")

weibodata.heading=weibodata.heading.fillna( "N" ) weibodata.addr=weibodata.addr.fillna( "N" )

weibodata.city=weibodata.city.fillna("N")

weibodata.user_province=weibodata.user_province.apply ( lambda x: '其他' if x=='NIL' else x)

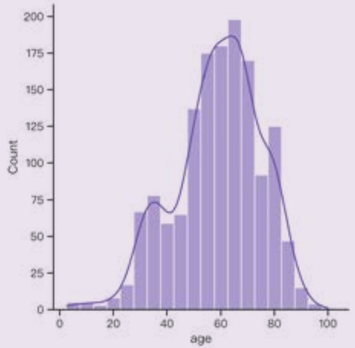
weibodata.pubdate=weibodata.pubdate.apply(lambda x: np.sum([int(t)*i for t,i in zip(x.split('/'),
[0,100,1])]))

if x[0]=='2' else 0) # Standardize the date format

clear_data=weibodata.copy( )[~((weibodata.content.isnull( ) )& (weibodata.heading.isnull() ))]
clear_data["content"] = clear_data["content"].apply( lambda para: re.findall(r"^(.+?)@",para)
[0].upper() if '@' in para else para.upper( ))

clear_data.head( )
```

PORTRAIT OF HELP SEEKERS

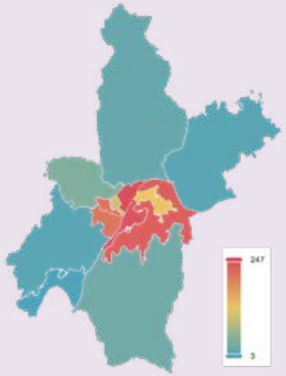


Age Description
The age of the seekers is primarily in the middle to elderly age group, with a predominant number of senior citizens. According to the curve, it is quite evident that this distribution has two peaks, with one minor peak around 30 years of age and the highest peak around 65 years of age.



Content Analysis
From the word cloud, it can be seen that 'Fever', 'Hospital' were highly mentioned. 'Father' and 'Mother' were also high-frequency world since a lot of posts were posted by people who lived far away from their elderly parents.

The number of help seekers

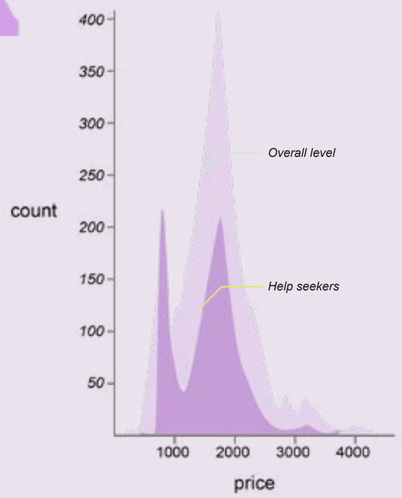


Help seekers' position distribution



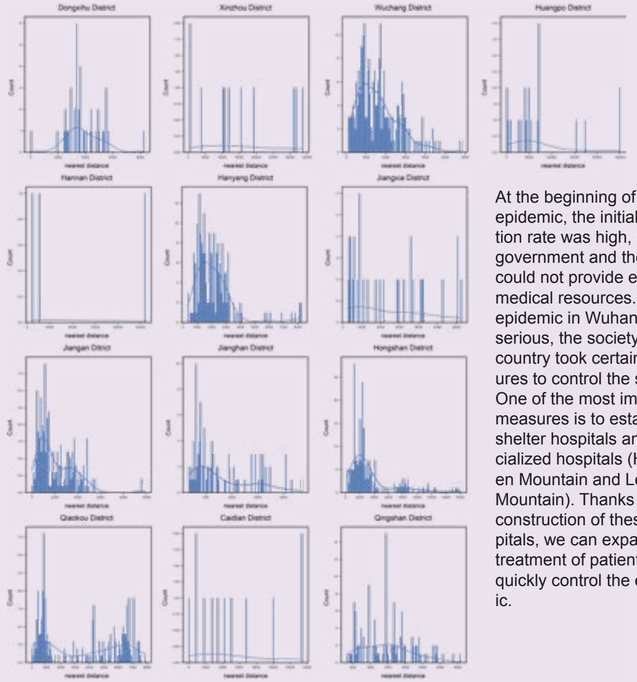
Location Analysis
Given all the longitude and latitude location data for all the help seekers and plotted their distribution in Wuhan city, and then obtain their district by calling Baidu API. It shows that most of them were located in the city central part.

Economy
We estimate the distribution of economical level of these help seekers through the house price they lived in, and compare it with the overall distribution of economical level. The price is obtained from a rental agency.



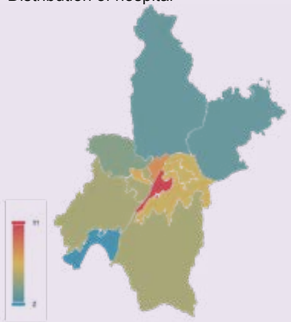
MEDICAL RESOURCES ANALYSIS

According to word frequency, we observe that most mentioned words are about hospital, fever, or other medical related words, which showed demands for medical help. To fully understand this phenomenon, we do a research on the medical resources in Wuhan as well as that surrounded these patients.



At the beginning of the epidemic, the initial infection rate was high, and the government and the country could not provide enough medical resources. As the epidemic in Wuhan became serious, the society and the country took certain measures to control the situation. One of the most important measures is to establish shelter hospitals and specialized hospitals (Huoshen Mountain and Leishen Mountain). Thanks to the construction of these hospitals, we can expand the treatment of patients and quickly control the epidemic.

Distribution of hospital



The distribution of hospitals in wuhan is not well-balanced and most of the hotels are concentrated in the central area.

On the right are the interactive timeline maps, which can show the information along with the time changing.





Number of patient

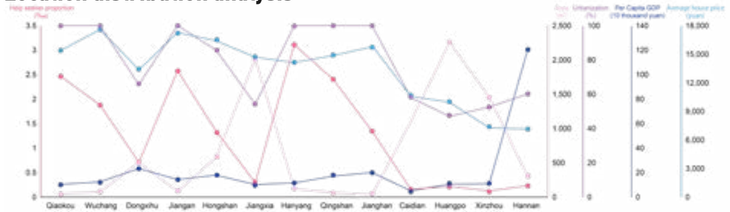
Number of hospital



A lot of patients had available medical resources but could not use because of the politics of covid-19 whereas a large proportion of patients were out of reach for that, especially in the places with less hospitals.

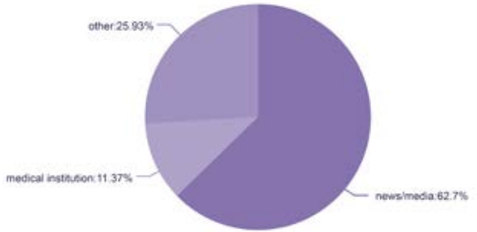
OTHER EXPLORATIONS

Location distribution analysis



This part uses some exterior data to explore the connection of the help seekers location distribution. To eliminate the influence of demographic differences, I focused on the proportion of individuals seeking assistance as our subject of observation. The chart reveals that the correlation of the proportion of help seekers and the economy is not significant. However, the proportion of help seekers has a strong positive correlation with urbanization rate of the population, and average housing prices, and a negative correlation with regional area.

Who to ask for ?



To increase the exposure of post, user always @ other users, which usually are the verified accounts. By analyzing the frequency of @ users, we can identify who users can easily seek help from in this event. We can see that news or media accounts takes up the large proportion in this group.

07 PUBLIC EMERGENCY DATA COLLECTION AND VISUALIZATION PLATFORM

The lockdown of Wuhan and the heavy rains in Henan are major public events that have caused heavy damage to our country in the past few years. They show the potential of pan-media, a brand-new response measure for public emergencies. It introduces ordinary people into the traditional official-led emergency mechanism, implements online rescue while on-site rescue by responsible departments, and can provide real-time feedback on rescue information. However, the "mutual aid ecosystem" automatically formed purely by social media still has flaws. For example, the data is redundant and is hard for people to extract useful information. Therefore, this project proposes to build a data platform model for collecting, processing, and visualizing public emergency information data from social media information and establish a prototype system.

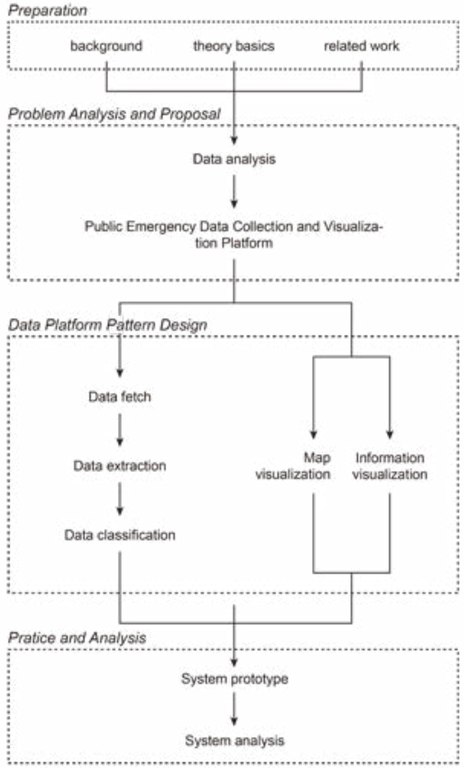


In July 2021, Zhengzhou and other areas in Henan Province experienced extremely rare and catastrophic floods due to sustained heavy rainfall.



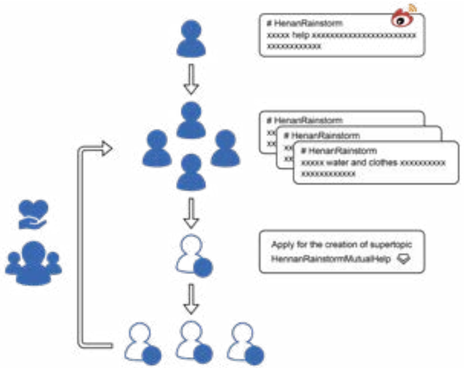
At the end of 2019, the COVID-19 epidemic began to emerge in China, and in early 2020, it completely broke out, ultimately sweeping the world. Wuhan, as the "epicenter" of the earthquake, began lockdown on January 23 and lasted for 76 days.

PROCESS



HENAN RAINSTORM DATA ANALYSIS

In order to study the phenomena we observed, verify the hypotheses we proposed, and design a suitable data platform model, we obtained the Weibo data from July 20, 2021 to July 22, 2021 by calling the API, and extracted useful information, and analyzed the obtained data after cleaning.

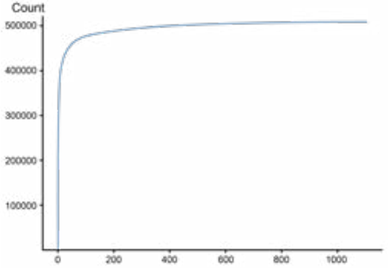


The formation process of the "mutual aid ecosystem" on social media: In the entire ecosystem, we can see that in addition to help-seekers, different roles such as information release organizers, information classification volunteers, and network rescuers have also been formed, with corresponding functions and tasks. This division of labor plays a significant impact in mutual aid.

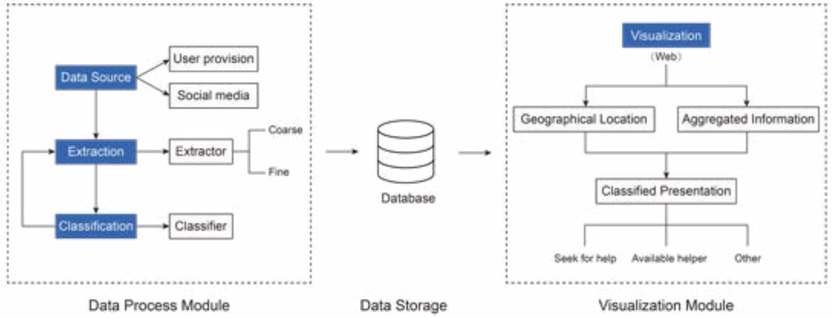


Word cloud: Through this chart, we can intuitively see what people cared about in this situation. There are qualifying words related to the crowd (Elderly people), problems and needs in the emergency (materials, lost contact), geographical location (Xinxiang, Gongyi City), and prayers and blessings (come on, be safe).

Posts forward distribution: The overall distribution showed a long-tail distribution which indicates the exist of Pareto principle. A small amount of information receives a large amount of exposure and it lead to the decrease the efficiency of the rescue and wastes a lot of time and human resources.



SYSTEM DESIGN



Data Collection
In this project, the data mainly comes from social media platforms. In theory, to ensure the timeliness of information, we need to update data in real-time, but updating data in real-time consumes a large amount of computing resources and storage resources. Many platforms also limit the frequency of crawling data. Therefore, to balance data timeliness and platform overhead, we adopt a batch processing mode to perform batch calculations on data.

Data Extraction
This function is mainly used to clean, integrate and extract information from the original data. After cleaning, the data must be integrated and structured information extracted to facilitate subsequent data visualization.

Data Classifications
Data classification includes a text classifier, which is used to classify roughly extracted text information into three categories: help-seeking posts, help-providing posts, and others.

Data Visualization
This part visualizes the obtained data to present the data more intuitively and conveniently. In public emergencies, the most important information includes geographical location information, contact information, and information content. Based on the above demands, we divide the visualization into geographical location and summary information. The former one is presented through a map, while latter is shown through a list and has filtering and retrieval functions.

REALIZATION

The prototype system adopts B/S architecture, which is the architectural model of browsers and servers. The system uses the open-source database MySQL as the back-end database, python is used on the server side, and the front-end of the system is implemented by HTML. During operation, the web server is responsible for receiving user requests and executing request responses, connecting to the database and submitting data processing requests to the database. The database then returns the processing results to the server, followed by the web server presenting the results to the user.

We use the Baidu API interface to determine its longitude and latitude based on the location information, and mark it based on the post classification, where red indicates seeking Help information, blue means providing help information, green means other. After clicking on the mark in the map, information data related to the location can be displayed to help users better understand the current situation in the area.





This module implements the presentation of all event-related information, in the form of tables. This part also implements user interaction functions. Users can input keywords to search, or filter based on information type (seeking help, providing help, and others) and release date.